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1. PROJECT OVERVIEW

1.1 PROJECT TITLE

CowBreeze: Smart Farm Solution for Rural-Urban Synergy via Wind-Customized Ammonia Scrubbing

1.2 KEY CHALLENGE AREAS

Urban Air Quality and Climate Resilience

1.3 SOLUTION

Mobile App

1.4 PROJECT SUMMARY

Ammonia gas from livestock farms can travel with the wind and contribute to fine dust pollution in urban areas[1, 2]. However, large-scale rural livestock farms often have little incentive to manage it due to high chemical costs[3].

Our system integrates NH_3 sensing and weather data to trigger directional ammonia scrubbing. When NH_3 exceeds the caution threshold under city-bound, spray-safe wind conditions, only the corresponding fence valve releases a fixed neutralizing dose. This converts ammonia into ammonium salt fertilizer[4, 5], reducing chemical costs, supporting eco-friendly subsidy compliance, and improving urban air quality[6] through urban-rural synergy.

2. CONTEXT AND PROBLEM STATEMENT

2.1 Environmental Context and Urgency

Ammonia emissions

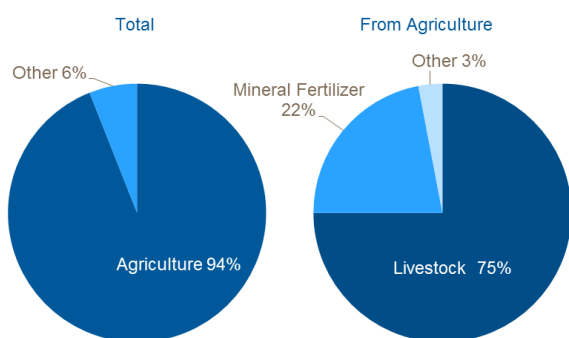


Figure 1. Ammonia Emissions in total (left) and from agriculture (right) (source: [7])

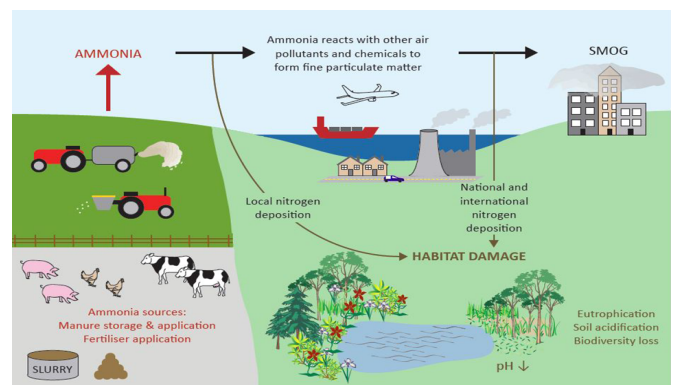


Figure 2. The environmental impacts of ammonia pollution. (source: [8])

Large-scale livestock farms emit substantial amounts of ammonia gas from manure daily (Figure 1). Carried by the wind over significant distances, these agricultural emissions combine with industrial and transport pollution in urban areas to form secondary fine particulate matter ($\text{PM}_{2.5}$). When inhaled, these background-level fine particles penetrate deeply into vital organs, contributing to severe cardiovascular and respiratory diseases[8].

Despite the critical nature of this environmental and public health threat, effective mitigation remains severely hindered in South Korea due to prohibitive economic barriers. According to domestic media reports[3], installing an essential ammonia reduction facility costs an estimated 1.2 billion KRW, with annual operational costs soaring up to 400 million KRW. Because rural farms face immense financial burdens, they have little economic incentive to consistently operate or invest in high-cost mitigation systems. Consequently, this market failure leaves a critical gap in controlling secondary $\text{PM}_{2.5}$ formation.

2.2 Scale

Approximately **8,500** large-scale beef cattle farms (owning over 100 heads) and **5,400** dairy farms nationwide[9].

2.3 Stakeholders

- **Wholesale Livestock Farms (Producers):** Located in blind spots with no complaints; want government subsidies but abandon systems due to high operational (chemical) costs.
- **Urban Residents (Victims):** Primary victims facing respiratory health risks from rural-born fine dust.
- **Government and Local Authorities (Regulators):** Waste budgets by distributing hardware without software management, failing to reduce actual urban $\text{PM}_{2.5}$ levels.

2.4 Negative Impacts and Consequences

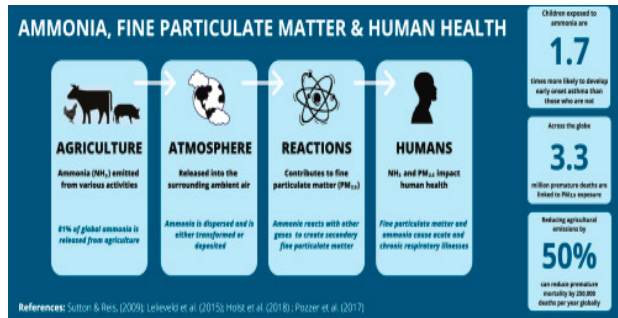


Figure 3. How Agricultural Ammonia Emissions Transform into Fine Particulate Matter ($PM_{2.5}$) and Impact Human Health. (source: [6])

- **Environmental and Health Damage:** Increased fine dust alerts and massive social medical costs for citizens.

- **Policy Failure:** Expensive environmental infrastructure is left unused, wasting national budgets and deepening rural-urban environmental conflicts.

3. EXISTING SOLUTIONS AND GAP ANALYSIS

3.1 Currently Available Solutions in the Market



Figure 4. Biofilter System(source: [10])

- **Large-scale Biofilters Systems (Figure 4):** Hardware systems designed to capture and filter ammonia gas inside the farm.

- **Manual Misting Systems:** Spraying devices that operate manually to spread neutralizing agents.

3.2 Limitations and Gap Analysis

- **Financial Barriers:** Biofilters require thousands of dollars for setup/maintenance, making adoption impossible for independent wholesale farms[3].
- **Chemical and Resource Waste:** Timer systems spray 24/7 regardless of weather, wasting expensive chemicals when there is no wind or when wind blows away from target zones.
- **The Missing Gap:** Lack of an intelligent software that minimizes operational costs through real-time, wind-based precision control

4. PROPOSED SOLUTION AND CORE FEATURES

Our solution is a **Wind-Customized Ammonia Mitigation Software** that integrates NH₃ sensor with Openweather API[11] information. When ammonia reaches a predefined threshold under city-bound wind conditions, the software automatically activates the fence valve facing the urban area to spray a fixed dose(60mL/m)[12] of neutralizing agent, preventing unnecessary chemical use while reducing ammonia dispersion.

Core Features

1. **Smart Spray & Direction Control:** Automatically controls spray timing and direction based on real-time NH₃ concentration, temperature, wind speed, and wind direction.
2. **Eco-Subsidy Data Logging:** Records spray operations and chemical usage, generating visualized reports to support government eco-friendly subsidy applications.

5. INNOVATION & TARGET GROUPS & POTENTIAL IMPACT

5.1 Innovation

• **Chemical Transformation:** Moves beyond simple odor masking. It combines real-time weather data with chemical scrubbing to transform airborne ammonia into solid ammonium salt a high-quality fertilizer component safely dropping it to the ground.

5.2 Target Audience

• **Wholesale (B2B) Livestock Farms Outside Cities:** Farms in regulatory blind spots that lack motivation for emission control but need to reduce chemical costs and receive government eco-subsidies.

5.3 Potential Impact

- **Environmental:** Drastically reduces urban fine dust ($PM_{2.5}$) by blocking ammonia on its pathway and promotes resource recycling through gas-to-fertilizer transformation.
- **Economic:** Maximizes farm profits by saving up to 70% on chemical costs while securing government subsidies; increases government budget efficiency.

- Social: Lowers respiratory medical costs for urban residents and resolves environmental conflicts between rural and urban areas.

6. DESCRIPTION OF TECHNOLOGIES APPLIED

6.1 Proposed Technologies

- Mobile Frontend: Swift(iOS)

6.2 System Architecture

The system adopts a rule-based environmental decision architecture that determines binary spray activation (ON/OFF) and directional valve selection. We combines real-time NH_3 sensor with Openweather API data[11] to trigger spraying only under predefined environmental conditions. The software control pipeline consists of three sequential decision modules.

① NH_3 Threshold Evaluation (G_{NH_3})

Since ammonia volatilization increases under higher temperatures, the trigger threshold is adjusted according to ambient temperature, enabling earlier responses during hot weather conditions.

② Wind Direction Evaluation (G_{dir})

Wind direction data obtained from the OpenWeatherMap API are compared with the predefined urban direction. If the wind is carrying ammonia toward the urban area, the corresponding fence valve is selected for spraying; otherwise, spraying is skipped.

③ Wind Speed Evaluation (G_{wind})

The system verifies whether the current wind speed falls within the stable spraying range (1.5-3.6 m/s). This operational window, based on EPA spray drift recommendations, prevents spraying during stagnant air or excessive wind conditions.

Final Spray Decision

Spraying is activated only when all three conditions are satisfied:

$$S = G_{\text{NH}_3} \cdot G_{\text{dir}} \cdot G_{\text{wind}} \quad \begin{cases} S = 1, \text{Spray} \\ S = 0, \text{No Spray} \end{cases}$$

6.3 Data and Infrastructure Requirements

- Data
 - OpenWeatherMap API: Current Weather API[11]

7. REFERENCES

- [1] UCL News. (2023, March 31). *Farms found to be the biggest particulate pollution source for cities*. UCL. <https://www.ucl.ac.uk/news/2023/mar/farms-found-be-biggest-particulate-pollution-source-cities>
- [2] Hristov, A. N. (2011). Technical note: Contribution of ammonia emitted from livestock to atmospheric fine particulate matter (PM_{2.5}) in the United States. *Journal of Dairy Science*, 94(6), 3130-3136. <https://doi.org/10.3168/jds.2010-3681>
- [3] *The Farmers Newspaper (Nongmin Sinmun)*, June 2024. <https://www.nongmin.com/article/20240605500781>
- [4] Hadlocon, L. J. S., Manuzon, R. B., & Zhao, L. (2015). Development and evaluation of a full-scale spray scrubber for ammonia recovery and production of nitrogen fertilizer at poultry facilities. *Environmental Technology*, 36(1-4), 405-416. <https://doi.org/10.1080/09593330.2014.950346>
- [5] Rietra, R., van Dijk, K., & Schoumans, O. (2024). Environmental Effects of Using Ammonium Sulfate from Animal Manure Scrubbing Technology as Fertilizer. *Applied Sciences (Switzerland)*, 14(12), Article 4998. <https://doi.org/10.3390/app14124998>
- [6] Wyer, K. E., Kelleghan, B. D., Blanes-Vidal, V., Schaubberger, G., & Curran, T. P. (2022). Ammonia emissions from agriculture and their contribution to fine particulate matter: A review of implications for human health. *Journal of Environmental Management*, 323, Article 116285. <https://doi.org/10.1016/j.jenvman.2022.116285>
- [7] Jacquet, S. (2018, June 4). *Reducing ammonia emissions from agriculture*. Yara Ireland. <https://www.yara.ie/crop-nutrition/agronomy-advice/reducing-ammonia-emissions/>
- [8] Department for Environment, Food & Rural Affairs. (2024, January 1). *Code of Good Agricultural Practice (COGAP) for Reducing Ammonia Emissions*. GOV.UK. <https://www.gov.uk/government/publications/code-of-good-agricultural-practice-for-reducing-ammonia-emissions/code-of-good-agricultural-practice-cogap-for-reducing-ammonia-emissions>
- [9] Statistics Korea. (2026). Number of Households and Head of Livestock by Province and Rearing Scale (Hanwoo, Beef Cattle) Available at: <https://kosis.kr/>
- [10] Kyeonggi Ilbo. (2026, June 9). "Pocheon City reduces ammonia from livestock waste treatment facilities by 85%... achievements in improving livestock odor". Daum News. <https://v.daum.net/v/bBYKlxx2yI>
- [11] https://openweathermap.org/api/current?collection=current_forecast
- [12] Kim, T.-H., Park, H.-Y., & Kim, S. (2000). Characteristics of ammonia removal by natural neutralizer. *Journal of Korean Society of Environmental Engineers*, 22(4), 651-658. [https://www.jksee.or.kr/upload/pdf/KSEE_1_2000_04_651\(C\).pdf](https://www.jksee.or.kr/upload/pdf/KSEE_1_2000_04_651(C).pdf)

